Perceived employability of senior executives

Introduction to Machine Learning - Individual project

Ali CAVUSOGLU

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1. Question and context

In France, executives over 45 who lose their job often struggle to find a new one. HR departments and outplacement firms want to know what makes a senior executive feel employable, and what distinguishes job seekers from employed executives. I use the answers of senior executives aged 45 to 65 to a questionnaire with 12 dimensions of perceived employability (Deacken & Jaotombo, 2019).

My first question is: which factors go with professional self-efficacy, the confidence in being able to find another job? This score is the closest to perceived employability. One of its questions is “I could easily obtain a job similar to mine in different types of organisations”. It also suits a linear regression: it is low on average (mean 3.55 on a 1 to 7 scale) but well spread (SD 1.47, cell 37).

My second question, central for outplacement firms, is: who are the job seekers? Job status is a yes/no outcome, so I use a logistic regression. Only 67 of the 457 respondents (14.7%) are job seekers (cell 36).

2. Data

The file has 460 respondents (cell 8). Each of the 12 scores is the mean of 3 to 7 questions on a 1 to 7 agreement scale. There are no missing values (cell 11). Three pairs of rows were identical on all 22 variables, including the 12 decimal scores (cell 15). I treated them as double submissions and dropped one row of each pair. The analysis uses 457 respondents (cell 16).

The sample is not random. The questionnaire was e-mailed to 1,515 senior executives, and 500 answered (questionnaire documentation). Most respondents come from one business school: 80.3% hold a business school degree and 67.4% earn more than 99k (Table 1 - cell 36; Figure 1 - cell 34). So the results describe this group only.

Table 2 sums up the data preparation (cell 90). I merged very small categories, because a category with few people gives an unstable coefficient. For example, construction (7 people, no job seeker, cell 65) joined industry. For salary, two thirds of the respondents are in the open top band, and mean self-efficacy is not a straight line (3.53, 3.18 and 3.66 in three groups, cell 78). Codes 1 to 10 would assume equal steps, so I used the three groups as dummies (0/1 variables). I dropped education place, because 98.5% studied in France (Table 1 - cell 36). Family situation and number of children overlapped and sometimes disagreed: 27 “Couple” respondents report dependent children (cell 17). So I kept one variable for living in a couple and one for the number of children.

Table 1 - Respondent profile, n = 457 (notebook Table 1, cell 36)

Variable	Category	n	%
age	mean 50.6 (SD 3.8), median 51, range 45-65	457	
sex	Male / Female	282 / 175	61.7 / 38.3
job_status	Employed / Job seeker	390 / 67	85.3 / 14.7
education_level	Bachelor / Master / Beyond master	11 / 176 / 270	2.4 / 38.5 / 59.1
degree_type	Business school / University / Engineering school / Other	367 / 35 / 34 / 21	80.3 / 7.7 / 7.4 / 4.6
education_place	France / Abroad	450 / 7	98.5 / 1.5
sector	Industry / Finance & insurance / Other / Business support services	101 / 91 / 79 / 55	22.1 / 19.9 / 17.3 / 12.0
	Trade, transport, hotels & restaurants / Information & communication	46 / 40	10.1 / 8.8
	Education, health & social work / Real estate / Construction	29 / 9 / 7	6.3 / 2.0 / 1.5
family_status	Couple with children / Single / Couple	336 / 63 / 58	73.5 / 13.8 / 12.7
children	0 / 1 / 2 / 3 / 4 or more	73 / 42 / 147 / 129 / 66	16.0 / 9.2 / 32.2 / 28.2 / 14.4
salary_band	< 20k / 20-29k / 30-39k / 40-49k / 50-59k	6 / 5 / 13 / 20 / 15	1.3 / 1.1 / 2.8 / 4.4 / 3.3
	60-69k / 70-79k / 80-89k / 90-99k / > 99k	23 / 23 / 22 / 22 / 308	5.0 / 5.0 / 4.8 / 4.8 / 67.4

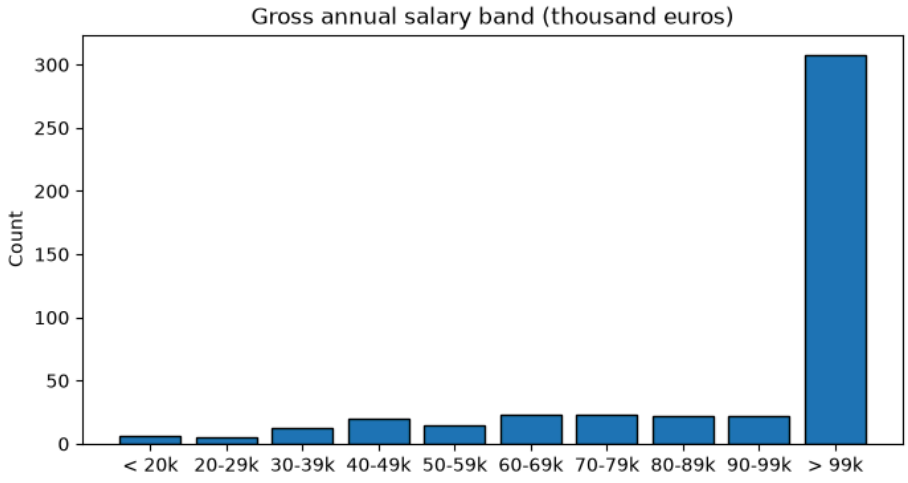


Figure 1 - Gross annual salary band (cell 34)

Table 2 - Data preparation decisions (notebook Table 7, cell 90)

Variable	Decision	Reason
duplicated rows	3 copies dropped (n = 457)	identical on all 22 variables
sector	Construction -> Industry, Real estate -> Finance & insurance (7 sectors)	7 and 9 people, Construction had no job seeker
education_level	Bachelor + Master -> Master or below	only 11 bachelors
degree_type	Business school vs other degree	80% business school, chi-square becomes valid
salary_band	3 groups as dummies (< 60k, 60-99k, > 99k)	67% in the open top band, pattern not linear
education_place	dropped	98.5% France, almost constant
family_status, children	couple (0/1) + children (0-4)	same information counted once
references	Male, Industry, Master or below, Business school, > 99k	largest or most natural group
job_status	job_seeker: 1 = job seeker, 0 = employed	outcome of Part 4

3. Exploratory analysis

Some scores are high for almost everyone (Table 3 - cell 37; Figure 2 - cell 38). Competence has a mean of 6.32, nobody scores below 4, and 36.76% give the maximum (Table 3 - cell 37). Few executives would say they were not competent in their job. Because of this ceiling effect, competence hardly separates respondents. Anticipation is also high (5.68, cell 37), because its questions describe what a good professional should think. Alternative employment is low (3.00, cell 37). Most of its questions are about stepping down, for example “I might accept a manual job”.

All 66 correlations between the scores are positive (Figure 3 - cell 40). Two groups go together: value awareness, market knowledge, flexibility and self-efficacy; and know-how visibility, networks and job search techniques. Seven pairs have r between 0.518 and 0.584 (cell 42), so some predictors will share information. I used Pearson correlations, which match the straight-line links of a regression (Spearman values differ by less than 0.04, cell 41).

I describe effect sizes with Cohen’s (1988) thresholds (for η^2: 0.01 small, 0.06 medium, 0.14 large). With about 100 tests, a few small p-values are expected by chance, so I look at effect sizes first. Personal variables are only weakly linked to the scores: no η^2 is large, and only two are medium, both for sector (cell 47). The only clear link with self-efficacy is age (r = -0.18, cell 45).

Job seekers score higher than employed executives on 10 of the 12 dimensions (Table 4 - cell 49). The largest gaps are training (+0.85; Cohen’s d = -0.661, cell 59) and networks (+0.65) (Table 4 - cell 49). But they are less confident: self-efficacy 3.23 against 3.61 (Table 4 - cell 49). Job seeking is also linked to having no dependent children (V = 0.238): 33% of those without children are job seekers, against 8% of those with four or more (cell 70). It is also linked to not having a business school degree (30.0% against 10.9%, V = 0.215, cells 75-76).

Table 3 - The 12 scores, n = 457 (notebook Table 2, cell 37)

Score	Mean	SD	Min	Median	Max	Skewness	% at 7	% at 1
competence	6.32	0.68	4.00	6.33	7.0	-0.75	36.76	0.00
anticipation	5.68	0.85	3.33	5.67	7.0	-0.39	10.28	0.00
job_search_techniques	5.44	1.12	2.00	5.75	7.0	-0.58	10.50	0.00
value_awareness	5.33	1.10	1.50	5.50	7.0	-0.46	9.19	0.00
flexibility	5.22	1.00	2.67	5.33	7.0	-0.01	7.22	0.00
market_knowledge	5.19	1.24	1.00	5.33	7.0	-0.70	10.94	0.66
versatility	5.09	1.23	1.00	5.00	7.0	-0.44	8.75	0.22
networks	4.96	1.52	1.00	5.20	7.0	-0.55	9.63	0.66
know_how_visibility	4.95	1.16	1.33	5.00	7.0	-0.19	7.00	0.00
training	3.96	1.32	1.29	3.71	7.0	0.40	1.97	0.00
self_efficacy	3.55	1.47	1.00	3.33	7.0	0.27	2.19	4.38
alternative_employment	3.00	1.36	1.00	2.86	7.0	0.44	0.44	6.78

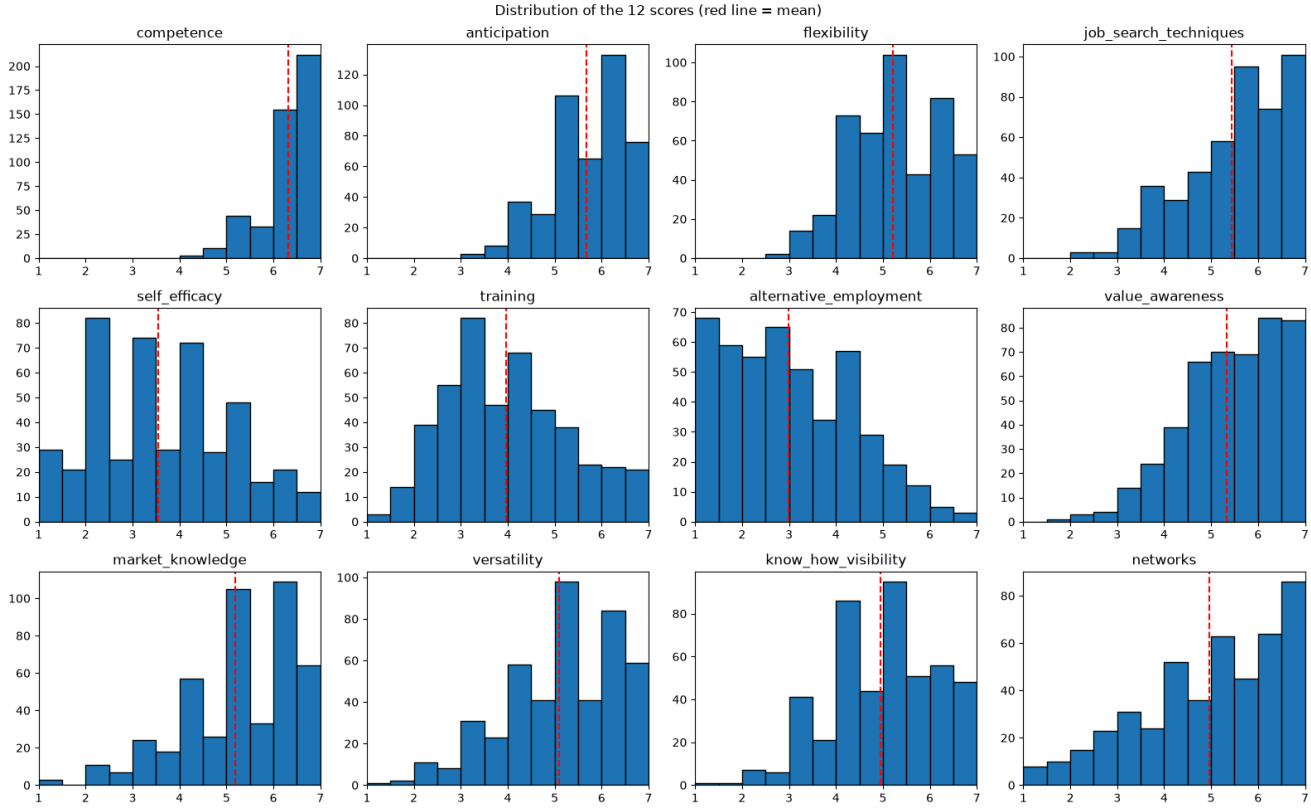


Figure 2 - Histograms of the 12 scores (cell 38)

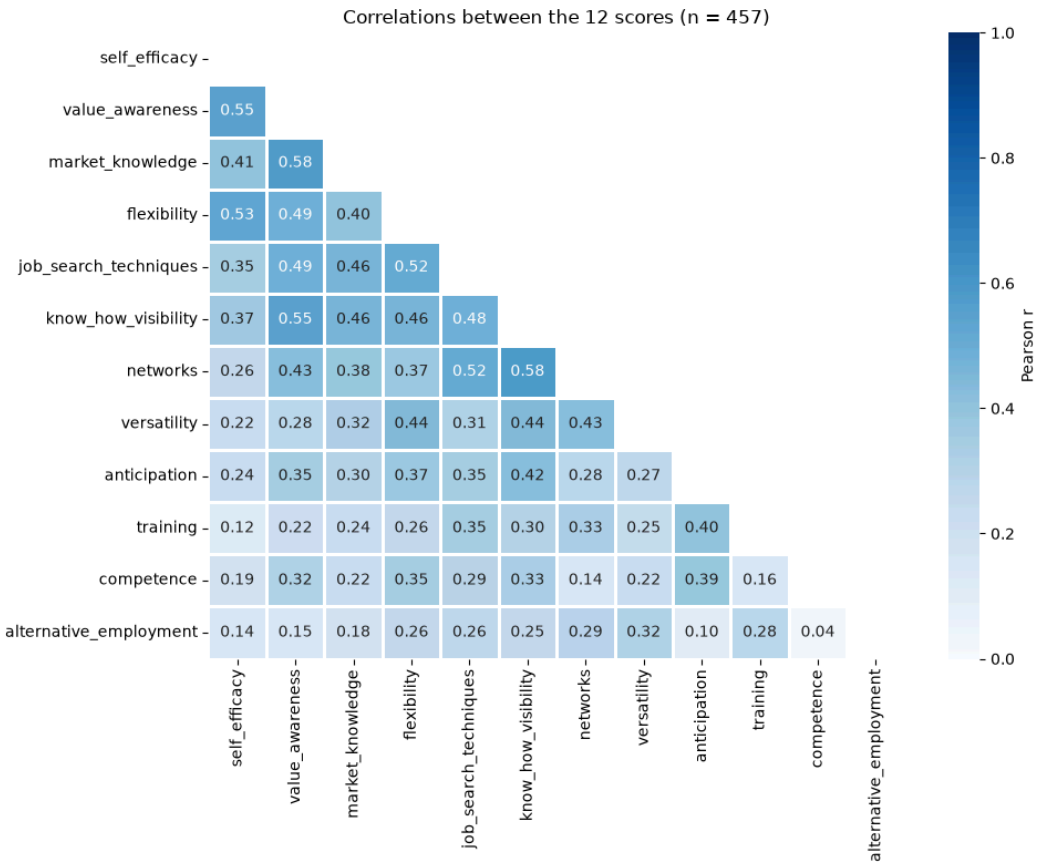


Figure 3 - Correlations between the 12 scores (cell 40)

Table 4 - Mean scores of employed people and job seekers (notebook Table 5, cell 49)

Score	Employed	Job seeker	Difference	η^2
training	3.83	4.69	0.85	0.05
networks	4.87	5.52	0.65	0.02
know_how_visibility	4.88	5.37	0.49	0.02
job_search_techniques	5.37	5.82	0.46	0.02
alternative_employment	2.93	3.38	0.45	0.01
market_knowledge	5.14	5.49	0.35	0.01
flexibility	5.17	5.51	0.34	0.01
versatility	5.05	5.37	0.33	0.01
anticipation	5.64	5.88	0.24	0.01
value_awareness	5.30	5.51	0.21	0.00
competence	6.32	6.36	0.04	0.00
self_efficacy	3.61	3.23	-0.37	0.01

4. Models

Linear regression. I fit self-efficacy by ordinary least squares (statsmodels). I use 25 predictors (cell 91): the 11 other scores and all the personal variables. Dropping weak variables after Part 1 would make the p-values look too good. I leave out job status: it is the outcome of the second model, and the direction of its link with confidence is unclear. Value awareness includes “I think my profile is likely to interest several potential employers”. Flexibility includes “It is easy for me to change company if I consider it necessary”. Both are close to the self-efficacy questions. I keep them because the scale has 12 separate dimensions and only one question

in each overlaps. The variance inflation factor (VIF) shows how much a predictor is explained by the others. All VIFs are at most 2.23 (cell 92), below the usual alarm level of 5 to 10 (James et al., 2023).

Logistic regression. With 67 job seekers (cell 101), a common rule of about 10 events per predictor (Peduzzi et al., 1996) allows few predictors. I keep the 12 scores and the personal variables linked to job status in Part 1: age, sex, children, couple and degree type. I leave out sector, education level and salary, which show no link with job status ($p = 0.538$, 0.875 and 0.287, cell 103). This gives 17 predictors (cell 104), with VIFs at most 2.29 (cell 105).

Evaluation. As in the course, the odds ratios come from a model fitted on all the data. To judge prediction, I split the data 80/20 into training and test sets, stratified by job status (random_state = 42, cell 110). A person is predicted to be a job seeker if the predicted probability is above 0.5. I compare the accuracy with a baseline that always predicts “employed”.

5. Results

The linear regression explains 44.8% of the differences in self-efficacy between respondents ($R^2 = 0.448$, cell 94). The adjusted R^2 (0.416, cell 94) is lower because it removes the small gain that any extra predictor brings by chance. Other things equal, one more point of flexibility is associated with 0.551 more points of self-efficacy, and one more point of value awareness with 0.449 more points (Table 5 - cell 95). Each extra year of age goes with 0.042 fewer points (cell 95). Compared with the > 99k group, the 60-99k group scores 0.354 lower and the < 60k group 0.416 lower (cell 95). Executives in education, health and social work score 0.548 higher than those in industry (cell 95). The other personal variables show no clear association.

Raw coefficients are hard to compare, because the predictors have different units. So I refit the model with all variables as z-scores (mean 0, standard deviation 1), which puts every coefficient on the same scale. Flexibility (0.374) and value awareness (0.335) are far ahead, followed by age (-0.107) and the two lower salary groups (-0.096 and -0.095) (cell 97; Figure 4 - cell 98). The residuals show no clear pattern and are close to normal (Figure 5 - cell 99).

An odds ratio multiplies the odds of being a job seeker, not the probability. The odds are job seekers per employed person, about 0.17 overall (67 for 390, cell 101). Other things equal, one more point of self-efficacy goes with 0.666 times the odds, and one more point of training with 1.352 times the odds (Table 6 - cell 108). Each extra dependent child goes with 0.688 times the odds, and a degree outside business school with 2.447 times (cell 108). Networks is not significant once the other scores are in the model (odds ratio 1.260, $p = 0.118$, cell 108).

Is the model useful? On the 92 test people (cell 110), accuracy is 0.8587, exactly the baseline (cell 112). The model flags only four people as job seekers, and two of them are right (Figure 6 - cell 113). Recall, the share of real job seekers that the model finds, is 0.15: 2 of 13 (cell 114). The model describes who the job seekers are, but it cannot find them.

Table 5 - Linear regression of self-efficacy, n = 457 (notebook Table 9, cell 95)

Predictor	Coef.	95% CI	p
const	1.241	[-0.709, 3.192]	0.212
competence	-0.118	[-0.296, 0.060]	0.193
anticipation	0.012	[-0.140, 0.164]	0.879
flexibility	0.551	[0.410, 0.691]	0.000
job_search_techniques	-0.012	[-0.140, 0.116]	0.850
training	-0.035	[-0.132, 0.061]	0.474
alternative_employment	0.036	[-0.053, 0.125]	0.422
value_awareness	0.449	[0.312, 0.587]	0.000
market_knowledge	0.107	[-0.003, 0.217]	0.057
versatility	-0.078	[-0.186, 0.029]	0.153
know_how_visibility	0.063	[-0.070, 0.197]	0.351
networks	-0.079	[-0.177, 0.019]	0.112
age	-0.042	[-0.071, -0.012]	0.006
children	0.072	[-0.023, 0.168]	0.138
couple	-0.083	[-0.430, 0.264]	0.640
sex_Female	0.066	[-0.172, 0.304]	0.588
sector: Business support services	0.195	[-0.192, 0.583]	0.322
sector: Education, health & social work	0.548	[0.056, 1.040]	0.029
sector: Finance & insurance	-0.155	[-0.473, 0.163]	0.339
sector: Information & communication	-0.311	[-0.736, 0.114]	0.151
sector: Other	0.021	[-0.324, 0.366]	0.906
sector: Trade, transport, hotels & restaurants	-0.295	[-0.694, 0.103]	0.146
education_level: Beyond master	-0.174	[-0.394, 0.046]	0.121
degree_type: Other degree	-0.019	[-0.319, 0.281]	0.903
salary: 60-99k	-0.354	[-0.652, -0.056]	0.020
salary: < 60k	-0.416	[-0.798, -0.033]	0.033

$R^2 = 0.448$, adjusted $R^2 = 0.416$ (cell 94). References: Male, Industry, Master or below, Business school, > 99k.

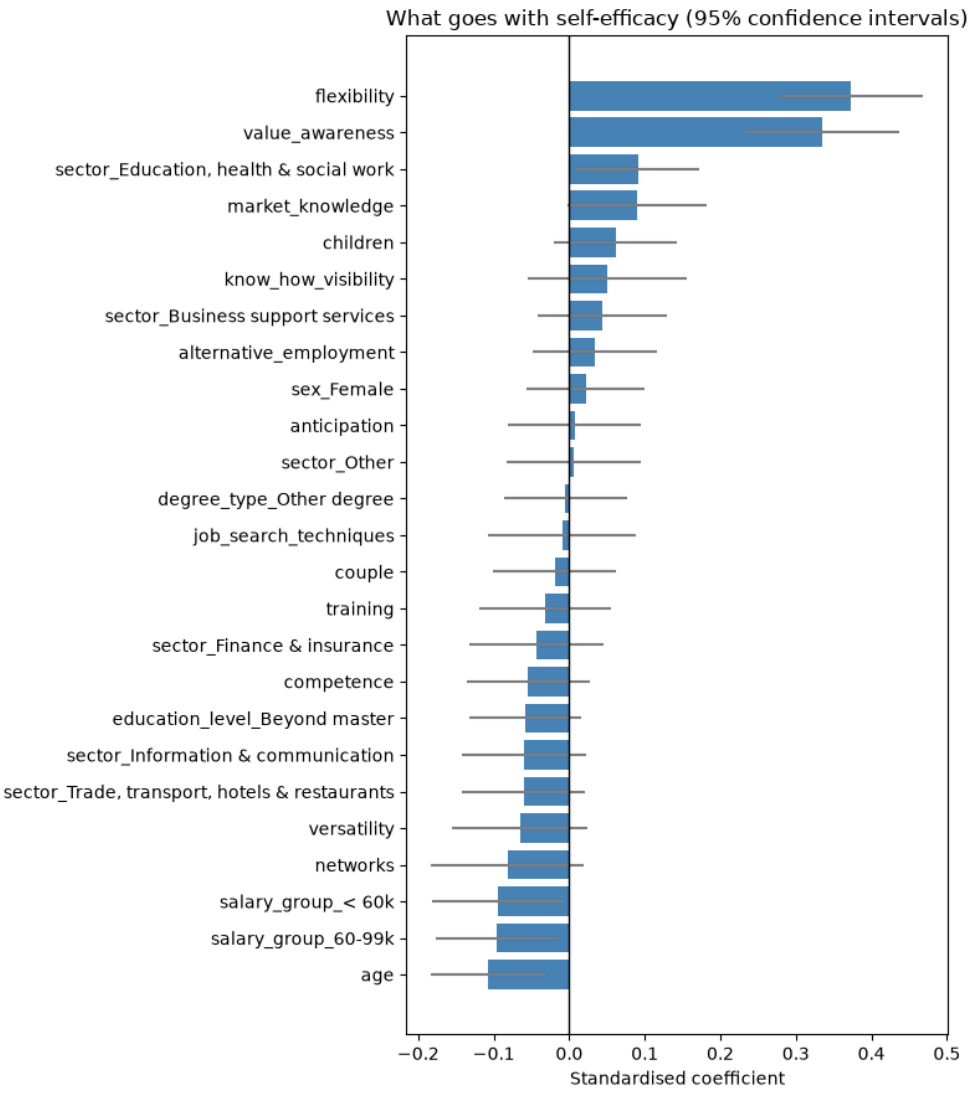


Figure 4 - Standardised coefficients with 95% confidence intervals (cell 98)

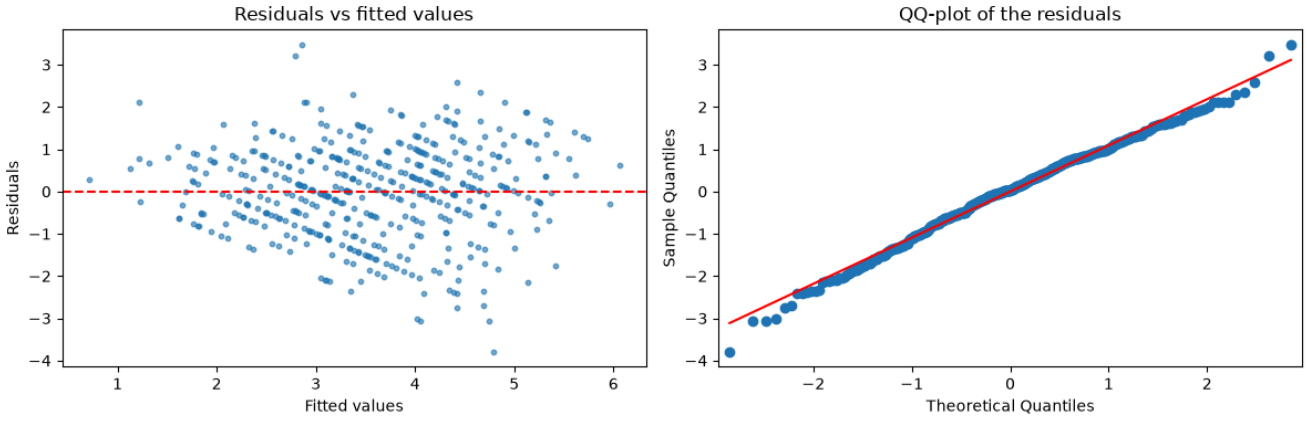


Figure 5 - Residuals vs fitted values and QQ-plot of the residuals (cell 99)

Table 6 - Logistic regression of being a job seeker: odds ratios, n = 457 (notebook Table 12, cell 108)

Predictor	Odds ratio	95% CI	p
competence	0.721	[0.438, 1.186]	0.197
anticipation	0.918	[0.590, 1.429]	0.705
flexibility	1.286	[0.856, 1.931]	0.225
job_search_techniques	1.045	[0.718, 1.520]	0.819
self_efficacy	0.666	[0.519, 0.855]	0.001
training	1.352	[1.045, 1.750]	0.022
alternative_employment	1.016	[0.807, 1.279]	0.894
value_awareness	1.137	[0.764, 1.694]	0.526
market_knowledge	1.099	[0.800, 1.509]	0.561
versatility	1.077	[0.792, 1.464]	0.637
know_how_visibility	1.138	[0.787, 1.646]	0.491
networks	1.260	[0.943, 1.683]	0.118
age	1.037	[0.961, 1.121]	0.349
sex_Female	1.285	[0.691, 2.389]	0.428
children	0.688	[0.531, 0.892]	0.005
couple	0.928	[0.402, 2.141]	0.861
degree_type: Other degree	2.447	[1.258, 4.758]	0.008

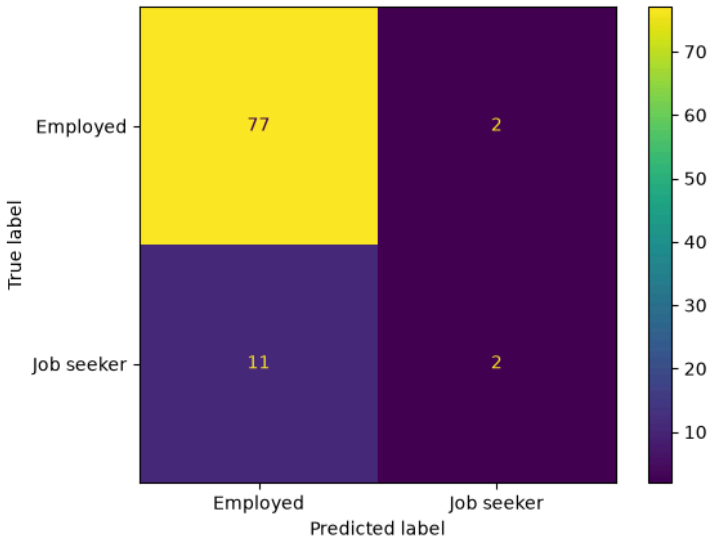


Figure 6 - Confusion matrix on the test set, 92 people (cell 113)

6. Limits

Ranking correlated predictors. The scores share information, and the model gives each shared part to only one of them. So five scores that are positively linked to self-efficacy on their own get a negative coefficient, for example job search techniques (r = 0.35) and networks (r = 0.26) (cell 100). Most confidence intervals include zero and overlap, and even the order of the top two could change in another sample. Low VIFs do not make this ranking reliable.

Content overlap. The two strongest predictors each contain one question close to the self-efficacy questions, so part of the R^2 may be circular.

Causality. Do job seekers train and network more because they look for a job, or the other way round? Some training questions are plans, such as “I plan to follow a training programme in the near future”. Planning to train is a natural reaction to a job loss, so here the job search probably comes first. Other training questions and all networks questions are about the past (“I have already used my network in the context of my job”), without saying if this was before or after the job loss. One survey at one moment shows associations, but not the direction of a link or its cause. A third factor, such as a redundancy with an outplacement programme, could explain both. Longitudinal data with the dates of training and job loss, or an experiment, would be needed.

Self-report. All scores are self-reported, and the very high competence scores suggest some answers reflect what people think they should say.

Few job seekers. The odds ratios are imprecise (about four job seekers per predictor). With 13 job seekers in the test set (cell 110), one person changes recall by about 8 points. A lower threshold than 0.5 would find more job seekers, but with more false alarms.

Sample and data. The sample is a convenience sample of mostly well-paid business school graduates. Family status and children sometimes disagreed, and the open top salary band hides differences above 99k.

7. Recommendations

These recommendations rest on associations in a self-reported, cross-sectional convenience sample, so they should be tested before wide use.

1. Help executives see their market value and their room to change company, for example with salary benchmarks and a review of their transferable skills. These two dimensions go most strongly with confidence (Table 5 - cell 95), even if part of this link comes from similar questions.
2. Give extra attention to older executives and to those earning less than 100k. They report lower confidence, other things equal (Table 5 - cell 95).
3. In outplacement, work on confidence, not only on training. Job seekers already train and network more, but they feel less confident, and lower confidence goes with higher odds of job seeking (Table 4 - cell 49; Table 6 - cell 108).
4. Do not use these scores to screen individuals for “job seeker risk”. The model finds only 2 of the 13 job seekers in the test set (Figure 6 - cell 113).
5. Follow executives over time, measuring confidence and job outcomes before and after training or outplacement.

References

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Peduzzi, P., Concato, J., Kemper, E., Holford, T. R., & Feinstein, A. R. (1996). A simulation study of the number of events per variable in logistic regression analysis. *Journal of Clinical Epidemiology*, 49(12), 1373-1379.

Appendix A - Traceability

Every number, table and figure comes from CAVUSOGLU_Ali_project.ipynb. A cell number is the In [n] number shown in the notebook after Restart & Run All. The report tables are numbered in reading order, so each table caption also gives its label in the notebook (for example, report Table 2 is notebook Table 7); the figures have the same numbers in both. The sample sizes of the survey (1,515 contacted, 500 answers) come from the questionnaire documentation.

Appendix B - AI use

Tools. Claude and Cursor, used as coding assistants in my project folder. The notebook ran on my own computer, and I did not share the data files.

How I used them. I planned the project phase by phase, following the parts of the brief. For each phase I gave the tools precise instructions: the task, the course methods to use, and my rules (reuse my helper functions without changes, and stop and wait for my choice before any critical change or decision). Within these instructions, they wrote and ran the notebook code and prepared first versions of the notes and of the report text. I then reviewed and edited them.

What I checked myself. I directed the analysis and took the main decisions myself, after asking for the options and their pros and cons. I dropped the three pairs of identical rows: two different people cannot give the same answers, to the last decimal, on 22 variables. I chose Pearson over Spearman because the scores are close to continuous and not strongly skewed, and because the regression in Part 3 looks for straight-line links; Spearman gave almost the same values, so the choice did not change the results. I decided how to merge the rare categories and to group salary into three bands instead of using codes from 1 to 10. I split family status and children into “couple” and “children”, because the two variables overlapped and sometimes disagreed. In the linear regression, I kept value awareness and flexibility even though one of their questions is close to self-efficacy, and I left job status out. In the logistic regression, I kept only 17 predictors, because 67 job seekers are too few for more. I reviewed the outputs and made sure that charts which added nothing to the analysis, such as the pairplot, were left out of the report and replaced by correct and useful ones.